

ALEXANDER VON MOLL

PERSONAL INFORMATION

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GOAL

Make meaningful research contributions to the area of cooperative control

WORK EXPERIENCE

	2017—	Research Engineer, AFRL/RQQA
Air Force Research Laboratory		Perform fundamental research as a member of the UAV Cooperative Control Team under the Autonomous Controls Branch of the Power and Control Division. Describe, define, and solve problems of interest to the Air Force in the areas of pursuit-evasion differential games and algorithms for persistent intelligence, surveillance, and reconnaissance.
	2012–2017	Research Engineer, AFRL/RQTE
Air Force Research Laboratory		Served as Integrated Product Team member under several turbine engine technology programs. Established plans and roadmaps for turbine engine control technologies. Supported research programs in turbine engine control software, actuators, and sensors.
	Sum. 2011	Software Development Intern, BOEING
Boeing		Developed support software and tools for the F-15SA flight control software group.

EDUCATION

	2019–2022	University of Cincinnati
PhD in Electrical Engineering		GPA: 4.0 · School: Engineering Research: <i>Skirmish-Level Tactics via Game-Theoretic Analysis</i> Advisor: Prof. Zachariah FUCHS
	2014–2015	Georgia Institute of Technology
Masters in Aerospace Engineering		GPA: 4.0 · School: Engineering Special Project: <i>Machine Learning Applications in Complex Control Systems</i> Description: The iterative linear quadratic Gaussian algorithm was re-derived, and the application of Gaussian Process Regression Networks to the algorithm was explored. Advisor: Assoc. Prof. Evangelos THEODOROU
	2008–2012	The Ohio State University
Bachelors in Aero/Astro Engineering		GPA: 3.97 · Minor: Computer Science · School: Engineering

Journal Articles

- Chapman, Thomas, Alexander Von Moll, and Isaac E. Weintraub. "Safe Navigation in the Presence of Range-Limited Pursuers". In: *Control Systems Letters* (May 26, 2026). DOI: [10.1109/LCSYS.2025.3645221](https://arxiv.org/pdf/2509.04258). URL: <https://arxiv.org/pdf/2509.04258>.
- Das, Goutam et al. "Dynamic Assignment Switching and Singular Surfaces in Multi-Agent Pursuit-Evasion Games". In: *Transactions on Automatic Control* (May 1, 2026). Submitted for Review.
- Surve, Prajakta, Shaunak D. Bopardikar, et al. "Heterogeneous Pursuit of an Active Target Under Sensing Constraints". In: *Control Systems Letters* (May 26, 2026). DOI: [10.1109/LCSYS.2025.3645033](https://arxiv.org/abs/2509.09777).
- Von Moll, Alexander, Dipankar Maity, et al. "Target Defense Using a Turret and Mobile Defender Team". In: *Control Systems Letters* (May 26, 2026). DOI: [10.1109/LCSYS.2025.3645833](https://arxiv.org/abs/2509.09777). URL: <https://arxiv.org/abs/2509.09777>.
- Bakker, Craig et al. "Operator-Theoretic Methods for Differential Games". In: *Journal of Computational Physics* (Apr. 1, 2025). Submitted for Review.
- Milutinović, Dejan, Alexander Von Moll, Satyanarayana G. Manyam, David W. Casbeer, Isaac E. Weintraub, and Meir Pachter. "Cooperative Pursuit-Evasion Games with a Flat Sphere Condition". In: *Open Journal of Control Systems* (Oct. 1, 2025). DOI: [10.1109/OJCSYS.2025.3604640](https://arxiv.org/abs/2405.07465).
- Milutinović, Dejan, Alexander Von Moll, Isaac E. Weintraub, et al. "Stochastic Optimal Avoidance of Multiple Engagement Zones". In: *Journal of Aerospace Information Systems* (May 1, 2025). DOI: [10.2514/1.I011593](https://arxiv.org/abs/2405.07465).
- Shishika, Daigo et al. "Deception in Turret Defense Game: Information Limiting Strategy to Induce Dilemma". In: *Open Journal of Control Systems* (Sept. 30, 2025). DOI: [10.1109/OJCSYS.2025.3611726](https://arxiv.org/abs/2405.07465). URL: <https://arxiv.org/abs/2405.07465>.
- Surve, Prajakta, Shaunak Bopardikar, et al. "Mutual Support by Sensor-Attacker Team for a Passive Target". In: *Open Journal of Control Systems* (Oct. 1, 2025). Accepted. DOI: [10.1109/OJCSYS.2025.3612246](https://arxiv.org/pdf/2509.06092). URL: <https://arxiv.org/pdf/2509.06092>.
- Weintraub, Isaac, Alexander Von Moll, David Casbeer, and Satyanarayana G. Manyam. "Virtual Target Selection for a Multiple-Pursuer-Multiple-Evader Scenario". In: *Journal of Aerospace Information Systems* (Jan. 22, 2025). DOI: [10.2514/1.I011510](https://arxiv.org/abs/2111.09205).
- Bajaj, Shivam, Shaunak Bopardikar, Eric Torng, et al. "Multi-vehicle Perimeter Defense in Conical Environments". In: *Transactions on Robotics and Automation* (Jan. 10, 2024). DOI: [10.1109/TRO.2024.3351556](https://arxiv.org/abs/2403.12346).
- Bajaj, Shivam, Bhargav Jha, et al. "Shortest Trajectory of a Dubins Vehicle with a Controllable Laser". In: *Transactions on Automatic Control* (Mar. 20, 2024). Accepted. URL: <https://arxiv.org/abs/2403.12346>.
- Dorothy, Michael et al. "One Apollonius Circle is Enough for Many Pursuit-Evasion Games". In: *Automatica* (Jan. 17, 2024). DOI: [10.1016/j.automatica.2024.111587](https://arxiv.org/abs/2111.09205). arXiv: 2111.09205 [math.OC]. URL: <https://arxiv.org/abs/2111.09205>.
- Jha, Bhargav et al. "Energy-Optimal Joint Motion Planning of a Pursuer-Turret Assembly". In: *Journal of Guidance, Dynamics, and Control* (Aug. 23, 2024). DOI: [10.2514/1.G008067](https://arxiv.org/abs/2401.14994). URL: <https://arxiv.org/abs/2403.14997v1>.
- Von Moll, Alexander and Meir Pachter. "Complete Solution of the Lady in the Lake Scenario". In: *Dynamic Games and Applications* (July 9, 2024). Presented at the International Symposium on Dynamic Games and Applications. DOI: [10.1007/s13235-024-00614-2](https://arxiv.org/abs/2311.06165). URL: <https://arxiv.org/abs/2401.14994>.
- Von Moll, Alexander and Isaac Weintraub. "Basic Engagement Zones". In: *Journal of Aerospace Information Systems* 21 (10 Sept. 12, 2024), pp. 885–891. DOI: [10.2514/1.I011394](https://arxiv.org/abs/2311.06165). URL: <https://arxiv.org/abs/2311.06165>.
- Bajaj, Shivam, Shaunak D. Bopardikar, et al. "Competitive Perimeter Defense with a Turret and Mobile Vehicle". In: *Frontiers in Control Engineering* (Feb. 27, 2023). DOI: [10.3389/fcteg.2023.1128597](https://arxiv.org/abs/2209.11289).
- Dillon, Patrick M. et al. "Optimal Trajectories for Aircraft Avoidance of Multiple Weapon Engagement Zones". In: *Journal of Aerospace Information Systems* 20 (8 June 12, 2023), pp. 520–525. DOI: [10.2514/1.I011224](https://arxiv.org/abs/2209.11289).
- Von Moll, Alexander, Zachariah Fuchs, Daigo Shishika, et al. "Turret Escape Differential Game". In: *Journal of Dynamics and Games* 11 (2 Sept. 1, 2023). Presented at the 19th ISDG., pp. 100–114. DOI: [10.3934/jdg.2023012](https://arxiv.org/abs/2209.11289). URL: <https://arxiv.org/abs/2209.11289>.
- Weintraub, Isaac, Alexander Von Moll, Eloy Garcia, et al. "Surveillance of a Faster Fixed-Course Target". In: *Transactions on Aerospace & Electronic Systems* (May 31, 2023). DOI: [10.1109/TAES.2023.3237129](https://arxiv.org/abs/2209.11289). URL: [http://arxiv.org/abs/2209.11289](https://arxiv.org/abs/2209.11289).
- Manyam, Satyanarayana Gupta, David W. Casbeer, et al. "Shortest Dubins Paths to Intercept a Target Moving on a Circle". In: *Journal of Guidance, Control, and Dynamics* (June 1, 2022). DOI: [10.2514/1.G005748](https://arxiv.org/abs/2209.11289).
- Von Moll, Alexander and Zachariah Fuchs. "A Lock-Evade, Engage or Retreat Game". In: *None* (Oct. 1, 2022). In preparation.
- Von Moll, Alexander, Meir Pachter, and Zachariah Fuchs. "Pure Pursuit with an Effector". In: *Dynamic Games and Applications* (Nov. 2, 2022), pp. 961–979. DOI: [10.1007/s13235-022-00481-9](https://arxiv.org/abs/2209.11289). URL: <https://arxiv.org/abs/2209.11289>.
- Von Moll, Alexander, Meir Pachter, Daigo Shishika, et al. "Circular Target Defense Differential Games". In: *Transactions on Automatic Control* 68 (7 Oct. 5, 2022), pp. 4065–4078. DOI: [10.1109/TAC.2022.3203357](https://arxiv.org/abs/2209.11289). URL: <https://arxiv.org/abs/2209.11289>.

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- Milutinović, Dejan, David W. Casbeer, et al. "Rate of Loss Characterization that Resolves the Dilemma of the Wall Pursuit Game Solution". In: *Transactions on Automatic Control* 68 (1 Dec. 27, 2021), pp. 242–256. DOI: [10.1109/TAC.2021.3137786](https://doi.org/10.1109/TAC.2021.3137786).
- Weintraub, Isaac E., Alexander Von Moll, Eloy Garcia, and Meir Pachter. "Maximum Observation of a Target by a Slower Observer in Three Dimensions". In: *Journal of Guidance, Control, and Navigation* 44 (3 Dec. 15, 2021). DOI: [10.2514/1.G005619](https://doi.org/10.2514/1.G005619).
- Garcia, Eloy, David Casbeer, et al. "Multiple Pursuer Multiple Evader Differential Games". In: *Transactions on Automatic Control* (May 1, 2020). DOI: [10.1109/TAC.2020.3003840](https://doi.org/10.1109/TAC.2020.3003840).
- Manyam, Satyanarayana G., David W. Casbeer, Alexander Von Moll, and Zachariah Fuchs. "Curvature Constrained Paths to Intercept a Target Moving on a Circle". In: *Transactions on Automation Science and Engineering* (Dec. 31, 2020). Submitted for Review.
- Pachter, Meir et al. "Cooperative Pursuit by Multiple Pursuers of a Single Evader". In: *Journal of Aerospace Information Systems* (Feb. 1, 2020). DOI: [10.2514/1.I010739](https://doi.org/10.2514/1.I010739). URL: <https://avonmoll.github.io/files/ManyPursuers.pdf>.
- Salmon, John L. et al. "Single Pursuer Multiple-Cooperative Evaders in the Border Defense Differential Game". In: *Journal of Aerospace Information Systems* (Mar. 19, 2020). DOI: [10.2514/1.I010766](https://doi.org/10.2514/1.I010766).
- Von Moll, Alexander, Eloy Garcia, et al. "Multiple Pursuer Single Evader Border Defense Differential Game". In: *Journal of Aerospace Information Systems* (Feb. 1, 2020). DOI: [10.2514/1.I010740](https://doi.org/10.2514/1.I010740). URL: <https://avonmoll.github.io/files/mp1e-border.pdf>.
- Pachter, Meir et al. "Two-on-One Pursuit". In: *Journal of Guidance, Control, and Dynamics* 42.7 (July 8, 2019). DOI: [10.2514/1.G004068](https://doi.org/10.2514/1.G004068). URL: <https://avonmoll.github.io/files/TwoCutters.pdf>.
- Von Moll, Alexander, David Casbeer, et al. "The Multi-Pursuer Single-Evader Game: A Geometric Approach". In: *Journal of Intelligent and Robotic Systems* 96 (2 Jan. 2, 2019), pp. 193–207. DOI: [10.1007/s10846-018-0963-9](https://doi.org/10.1007/s10846-018-0963-9). URL: <https://avonmoll.github.io/files/mp1e-journal.pdf>.
- Von Moll, Alexander, Meir Pachter, Eloy Garcia, et al. "Robust Policies for a Multiple Pursuer Single Evader Differential Game". In: *Dynamic Games and Applications* (10 May 4, 2019), pp. 202–221. DOI: [10.1007/s13235-019-00313-3](https://doi.org/10.1007/s13235-019-00313-3). URL: <https://avonmoll.github.io/files/mp1e-robust.pdf>.

Conference Papers

- Nimmagadda, Aditya et al. "Information-Theoretic Multi-Agent Path Planning for Risk-Aware Navigation Under Environmental Uncertainty". In: *Aviation Forum*. Submitted for Review. San Diego, USA: AIAA, June 8, 2026.
- Shah, Sagar, Alexander Von Moll, and Isaac Weintraub. "Strategic Evasion Using a Multi-Hypothesis Control Barrier Function". In: *Aviation Forum*. Submitted for Review. San Diego, USA: AIAA, June 8, 2026.
- Subramanian, Avinash et al. "Certification of Dynamical Systems using End-to-End Distributionally-Robust Uncertainty Quantification". In: *AIAA SciTech* 2026. Orlando, USA: AIAA, Jan. 12, 2026. DOI: [10.2514/6.2026-2022](https://doi.org/10.2514/6.2026-2022).
- Von Moll, Alexander and Isaac Weintraub. "Reactive Vehicle Guidance using Dynamic Maneuvering Cue". In: *AIAA SciTech*. Orlando, USA: AIAA, Jan. 12, 2026. DOI: [10.2514/6.2026-1199](https://doi.org/10.2514/6.2026-1199). URL: <https://arxiv.org/abs/2512.09083>.
- Chapman, Thomas, Isaac Weintraub, et al. "Engagement Zones for a Turn Constrained Pursuer". In: *International Conference on Unmanned Aerial Systems*. Accepted. Charlotte: IFAC, May 14, 2025, pp. 763–768. DOI: [10.1109/ICUAS65942.2025.11007894](https://doi.org/10.1109/ICUAS65942.2025.11007894).
- Denton, Alexander W. et al. "The Basic Engagement Zone in 3-D". In: *National Aerospace Electronics Conference*. Accepted. Dayton, USA: IEEE, July 28, 2025.
- Hansen, Alexander et al. "An Optimal Strategy for Off-Board Proximal Sensing of a Target: Part 2". In: *SciTech*. Orlando, FL: AIAA, Jan. 8, 2025. DOI: [10.2514/6.2025-1350](https://doi.org/10.2514/6.2025-1350).
- Manyam, Satyanarayana G., David W. Casbeer, Alexander Von Moll, and Isaac Weintraub. "Dubins Path with Terminal Range and Field-of-View Constraints". In: *Conference on Decision and Control*. Rio de Janeiro: IEEE, Dec. 9, 2025. DOI: [10.1109/LCSYS.2025.3582622](https://doi.org/10.1109/LCSYS.2025.3582622).
- "Shortest Dubins Path to a Moving Circle with Free Final Heading". In: *American Control Conference*. Accepted. Denver, CO: IFAC, July 8, 2025.
- Milutinović, Dejan, Alexander Von Moll, Isaac E. Weintraub, et al. "Policy for Optimal Avoidance of Multiple Engagement Zones". In: *SciTech*. Orlando, FL: AIAA, Jan. 8, 2025. DOI: [10.2514/6.2025-1544](https://doi.org/10.2514/6.2025-1544).
- Mora, Braulio, Animesh Chakravarthy, et al. "Influence of Malicious Agents in a Team Seeking to Contain an Evader". In: *SciTech*. Orlando: AIAA, Jan. 3, 2025. DOI: [10.2514/6.2025-1348](https://doi.org/10.2514/6.2025-1348).
- Surve, Prajakta, Richard L. Frost, et al. "Heterogeneous Pursuit of Multiple Translating Targets". In: *Conference on Control Technology and Applications*. San Diego: IEEE, Aug. 25, 2025. DOI: [10.1109/CCTA53793.2025.11151369](https://doi.org/10.1109/CCTA53793.2025.11151369).
- Von Moll, Alexander, Dejan Milutinović, et al. "One-vs-one Threat-Aware Weaponing with Basic Engagement Zones". In: *International Conference on Unmanned Aerial Systems*. Accepted. Charlotte: IFAC, May 14, 2025, pp. 1138–1145. DOI: [10.1109/ICUAS65942.2025.11007913](https://doi.org/10.1109/ICUAS65942.2025.11007913). URL: <http://arxiv.org/abs/2503.09475>.

- Weintraub, Isaac, Alexander Von Moll, David Casbeer, Satyanarayana G. Manyam, et al. "Min-Time Escape of a Dubins Car from a Polygon". In: *American Control Conference*. Accepted. Denver, CO: IFAC, July 8, 2025. URL: <https://arxiv.org/abs/2410.01589>.
- Weintraub, Isaac, Alexander Von Moll, Alexander Hansen, et al. "An Optimal Strategy for Off-Board Proximal Sensing of a Target: Part 1". In: *SciTech*. Orlando, FL: AIAA, Jan. 8, 2025. DOI: [10.2514/6.2025-1349](https://doi.org/10.2514/6.2025-1349).
- Wolek, Artur, David Casbeer, et al. "Maximum Kinetic Energy Paths for a Dubins Vehicle with Decaying Speed". In: *SciTech*. Orlando, USA: AIAA, Jan. 12, 2025. DOI: [10.2514/6.2025-1351](https://doi.org/10.2514/6.2025-1351).
- Maity, Dipankar et al. "Optimal Evasion From a Sensing Limited Pursuer". In: *American Control Conference*. Accepted. Toronto, Canada, July 8, 2024.
- Milutinović, Dejan, Alexander Von Moll, Satyanarayana G. Manyam, David W. Casbeer, Isaac E. Weintraub, and Meir Pachter. "Pursuit-Evasion on a Sphere and When it Can Be Considered Flat". In: *Conference on Decision and Control*. Milan, Italy: IEEE, Dec. 16, 2024. DOI: [10.1109/CDC56724.2024.10885887](https://doi.org/10.1109/CDC56724.2024.10885887). URL: <https://arxiv.org/abs/2403.15188>.
- Mora, Braulio, Alexander Von Moll, et al. "Escape from an Orbiting Pursuer with a Nonzero Capture Radius". In: *National Aerospace Electronics Conference*. Dayton, OH, July 8, 2024.
- Von Moll, Alexander, David W. Casbeer, Isaac E. Weintraub, et al. "Pure Pursuit of a Target on a Circular Trajectory". In: *AIAA SciTech*. Accepted. Orlando, FL, Jan. 30, 2024. DOI: [10.2514/6.2024-0956](https://doi.org/10.2514/6.2024-0956). URL: <https://isaacew.com/publication/vonmoll2024pure/vonmoll2024pure.pdf>.
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- Weintraub, Isaac E., Alexander Von Moll, David W. Casbeer, et al. "Virtual Target Selection for a Multiple-Pursuer Multiple-Evader Scenario". In: Orlando, FL, Jan. 30, 2024. DOI: [10.2514/6.2024-0123](https://doi.org/10.2514/6.2024-0123). URL: <https://arxiv.org/abs/2305.19399>.
- Weintraub, Isaac E., Alexander Von Moll, and Meir Pachter. "Minimum Time Escape from a Circular Region of a Dubins Car". In: *National Aerospace Electronics Conference*. Dayton, OH: IEEE, July 8, 2024. URL: <https://arxiv.org/abs/2405.05725>.
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- Weintraub, Isaac, Alexander Von Moll, Christian Carrizales, et al. "An Optimal Engagement Zone Avoidance Scenario in 2-D". In: *Scitech*. San Diego: AIAA, Jan. 3, 2022. DOI: [10.2514/6.2022-1587](https://doi.org/10.2514/6.2022-1587). URL: <https://arxiv.org/pdf/2111.05904.pdf>.
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OTHER INFORMATION

Awards & Recognition

- 2024 · Air Force Research Laboratory Jack Blackhurst Innovation Award
- 2022 · Excellent Reviewer – AIAA Journal of Guidance, Control, and Dynamics
- 2022 · University of Cincinnati Department of Electrical Engineering & Computer Science Outstanding Doctoral Dissertation Award
- 2019 · Aerospace Control & Guidance Systems Committee Dave Ward Memorial Lecture Award
- 2014 · AFRL Turbine Engine Division Civilian of the Year (for Excellence in Primary Job Duty)
- 2014 · DoD SMART Scholarship
- 2011 · DoD SMART Scholarship

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